

Tuesday, July 7, 2026 2:30 PM

Binomial RV

Hypergeometric RV

Poisson RV

Binomial RV

Toss a Fair Coin 5 times

$$P(\text{Exactly 3 Heads}) = \binom{5}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2$$

HHHTT
 $\left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2$

$$P(X=x) = p(x) = \binom{n}{x} \pi^x (1-\pi)^{n-x}$$

n independent trials

 π : p(success) on a trial

X: # of successes

$$E(X) = n\pi$$

$$\text{Var}(X) = n\pi(1-\pi)$$

$$\sigma_X = \sqrt{n\pi(1-\pi)}$$

$$X = Y_1 + Y_2 + \dots + Y_n$$

Y	P(Y)
0	$1-\pi$
1	π

$$E(Y) = (0)(1-\pi) + (1)(\pi) = \pi$$

$$\begin{aligned} E(X) &= E(Y_1 + \dots + Y_n) \\ &= E(Y_1) + E(Y_2) + \dots + E(Y_n) \\ &= n\pi \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= ? \quad \left[\begin{aligned} \text{Var}(Y) &= (0)^2(1-\pi) \\ &+ (1)^2(\pi) \\ &- \pi^2 \\ &= \pi - \pi^2 = \pi(1-\pi) \end{aligned} \right] \end{aligned}$$

$$= n\pi(1-\pi)$$

$$\begin{aligned} \text{Var}(X) &= \text{Var}(Y_1 + Y_2 + \dots + Y_n) \stackrel{\text{IND}}{=} \text{Var}(Y) \\ &= n\pi(1-\pi) \end{aligned}$$

The Binomial Distribution – Example

- A salesperson is about to visit 25 potential customers; the probability of a successful visit (a “sale”) is 0.3, independent of other visits. Let X be the number of sales.

- $p(8) = P(X = 8) = ?$ $\binom{25}{8} \cdot 0.3^8 \cdot 0.7^{17}$
- $\text{CDF}(8) = P(X \leq 8) = ?$ $p(0) + p(1) + \dots + p(8)$
- $E(X) = \mu_X = ?$ $(25)(0.3)$
- $\text{SD}(X) = \sigma_X = ?$ $\sqrt{(25)(0.3)(0.7)}$

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$$\text{Defn: } \text{CDF}(x) = P(X \leq x)$$

Jar with 20 marbles

12 Red

8 Green

Randomly select 5 marbles
w/ Replacement

$$P(\text{Exactly 3 Red}) = \binom{5}{3} .6^3 .4^2$$

Jar with 20 marbles

12 Red

8 Green

Randomly select 5 marbles

w/o Replacement

$$P(\text{Exactly 3 Red}) = \frac{\binom{12}{3} \binom{8}{2}}{\binom{20}{5}}$$

Hypergeometric RV : X

N : # objects in population

n : # objects in sample

r : # successes in population

x : # successes in sample

$$p(x) = \frac{\binom{r}{x} \binom{N-r}{n-x}}{\binom{N}{n}}$$

The Poisson Distribution – Example

- Phone calls arrive randomly to a call center at a rate of 4 calls per hour. Under the Poisson assumption, find
 - $P(5 \text{ calls between } 8 \text{ and } 10) = ?$
 - $P(0 \text{ calls between } 8 \text{ and } 8:15) = ?$
 - $P(\text{At least } 1 \text{ call between } 9 \text{ and } 10) = ?$

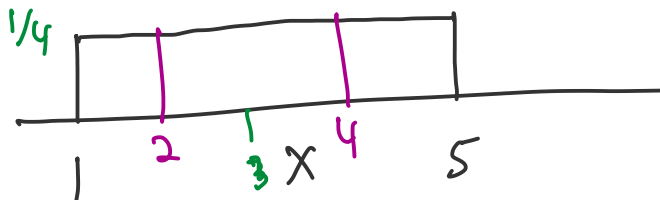
$$1. \lambda = 8 \text{ per } 2 \text{ hours} \quad \frac{e^{-8} 8^5}{5!}$$

$$2. \lambda = 1 \text{ per } \frac{1}{4} \text{ hour} \quad \frac{e^{-1} 1^0}{0!} = e^{-1}$$

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$$3. \lambda = 4 \text{ per hour} \quad 1 - \frac{e^{-4} 4^0}{0!} = 1 - e^{-4}$$

Continuous Random Variables



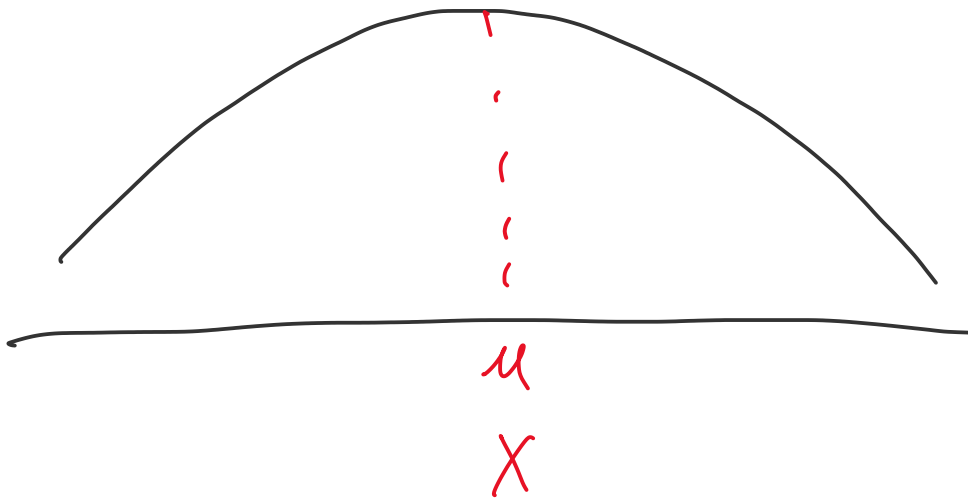
$$P(X=2) = 0$$

$$P(2 \leq X \leq 4) = \frac{1}{2}$$

$$P(1 \leq X \leq 2) = \frac{1}{4}$$

Normal Distribution

$$X \sim N(\mu, \sigma^2)$$



$$X \sim N(100, 10^2)$$

$$1. P(X < 112) = P\left(\frac{X - \mu}{\sigma} < \frac{112 - 100}{10}\right) = P(Z < 1.2) =$$

$$2. P(X > 83) = P\left(\frac{X - \mu}{\sigma} > \frac{83 - 100}{10}\right) = P(Z > -1.7) = 1 -$$

$$3. P(92 < X < 104) : \text{See below}$$

If $X \sim N(\mu, \sigma^2)$ then let's consider

$$Z = \frac{X - \mu}{\sigma}$$

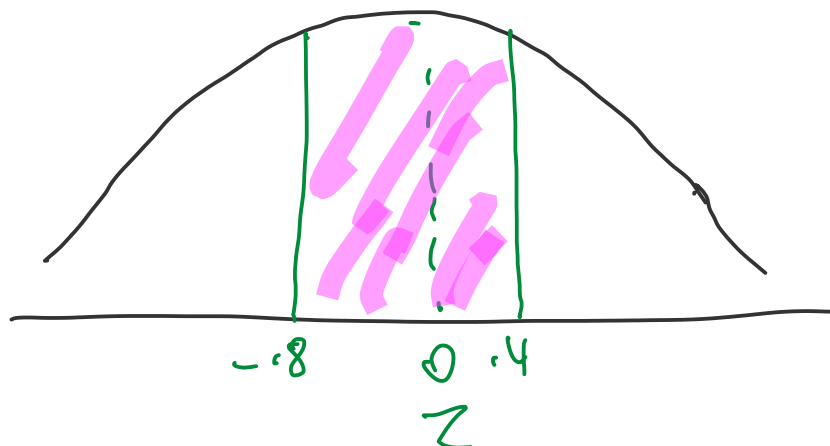
$$E(Z) = E\left[\frac{X - \mu}{\sigma}\right] = \left(\frac{1}{\sigma}\right) E[X - \mu] = \left(\frac{1}{\sigma}\right) 0 = 0$$

$$\text{Var}(Z) = \text{Var}\left[\frac{X - \mu}{\sigma}\right] = \left(\frac{1}{\sigma}\right)^2 \text{Var}(X - \mu) = \left(\frac{1}{\sigma}\right)^2 \sigma^2 = 1$$

$$Z \sim N(0, 1)$$

3 From Above

$$\begin{aligned} P(92 < X < 104) &= P\left(\frac{92 - 100}{10} < \frac{X - \mu}{\sigma} < \frac{104 - 100}{10}\right) \\ &= P(-.8 < Z < .4) \end{aligned}$$



$$= P(Z < .4) - P(Z < -.8)$$

$$= .6554 - .2119 = .4435$$